

PS2 NLG Assignment - Solution Coverage & Gap Analysis Report

Objective:

Validate whether uploaded solution files satisfy the requirements of Problem Statement 2: Controlled Natural Language Generation - Decoder vs LLM for Slot-Conditioned Response Generation.

Overall Finding:

The uploaded files are related to the assignment. Module 1 implementation is strong, covering dataset generation, schema design, linearization and vocabulary creation. However Model Development and Evaluation modules are missing.

Current Files:

- Assignment_PS2_NLG.pdf - Problem Statement
- 01_data_generation.ipynb - Dataset generation
- 02_data_validation.ipynb - Dataset checks
- 03_linearization.ipynb - Decoder input preparation
- 04_vocab_creation.ipynb - Token vocabulary creation
- schema.json
- train.json / validation.json / test.json
- linearized train/validation/test json
- vocab.json, token_to_id.json, id_to_token.json

Module Assessment:

Module 1: Data Preparation (3 Marks)

Task 1 Dataset Construction:

Status: Completed

- Retail delivery customer service domain created
- Multiple intents and slots available
- 400 samples generated
- Train/Validation/Test split available
- Edge cases covered:
missing slots, null values, empty values, duplicate entities, multiple products

Estimated Score: 1.4 / 1.5

Task 2 Linearization and Tokenization:

Status: Mostly Completed

- BOS/EOS tokens implemented
- Separator based linearization implemented
- Vocabulary generated

Missing:

- LLM prompt templates

Estimated Score: 1.2 / 1.5

Module 2: Model Development (3.5 Marks)

Task 3 Decoder Transformer:
Status: Missing

Required:
- Decoder-only Transformer or Seq2Seq model
- Training pipeline
- Loss calculation
- Inference generation

Need Add:
05_decoder_transformer.ipynb

Estimated Current Score: 0.2 / 2

Task 4 LLM Generation:
Status: Missing

Required:
- Prompt based LLM generation
- Test set inference

Need Add:
06_llm_generation.ipynb

Estimated Current Score: 0 / 1.5

Module 3: Evaluation and Analysis (3.5 Marks)

Task 5 Metrics:
Status: Missing

Required:
- Slot Accuracy
- BLEU
- ROUGE
- Hallucination Rate
- Latency
- Cost comparison

Need Add:
07_evaluation_metrics.ipynb

Estimated Current Score: 0 / 2

Task 6 Comparative Analysis:
Status: Missing

Required:

- Decoder vs LLM comparison table
- 10 generated examples
- Error diagnosis
- Deployment recommendation

Need Add:

08_final_analysis_report.ipynb

Estimated Current Score: 0 / 1.5

Final Estimated Score:

Current Submission: Around 2.8 / 10

Files Needed For 9+/10:

1. 05_decoder_transformer.ipynb
2. 06_llm_generation.ipynb
3. 07_evaluation_metrics.ipynb
4. 08_final_analysis_report.ipynb
5. decoder_predictions.json
6. llm_predictions.json
7. comparison_results.csv
8. README.md

Recommendation:

Do not submit current state. Dataset pipeline is strong, but model implementation and comparative evaluation

carry majority marks. Completing missing notebooks can move the assignment towards full marks.